

CSIS 4495 – Applied Research Project

Progress Report 4

GREENPATH – Sustainable Products Website with AI Chatbot Integration

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Work Date/Hours Log

The table below shows summary of the work done till date with the respective time spent on each task.

Date	Number of Hours	Description of Work Done
2025-10-23	3	Rearranged layout of Analytics page, moved the visualizations above the overall ranking table for better readability.
2025-10-24	3	Adjusted component structure to enhance rendering flow and reduce overlapping between charts and ranking tables.
2025-10-25	3	Fixed scroll behavior on Chat page: chat window now auto-adjusts to message height without screen jump during chat
2025-10-26	3	Enhanced Chatbot response alignment with the dataset; improved prompt structure for dataset-specific answers.
2025-10-27	3	Implemented forecasting logic in AnalyticsML.tsx using linear regression approximation for emission trends.
2025-10-28	3	Developed anomaly detection feature under AnalyticsML to identify irregular data patterns.
2025-10-29	3	Added predictive analysis module in AnalyticsML.tsx for country and sector-based emissions.
2025-10-30	3	Implemented clustering visualization under AnalyticsML.tsx to group countries by emission characteristics.
2025-10-31	3	Separated AnalyticsML from Analytics page; configured new header link and route for direct navigation.
2025-11-01	2	Conducted final integration test of AnalyticsML components and ensured charts render responsively.
2025-11-02	2	Updated Tailwind and layout alignment to maintain consistent styling across Analytics and AnalyticsML.
2025-11-03	2	Validated Chatbot output consistency against sample dataset queries.
2025-11-04	2	Performed bug testing, verified navigation links, and ensured visual consistency across pages.
2025-11-05	2	Reviewed overall UI integration, tested ML features, and finalized content for Report 4 submission.

Description of Work Done

In the past weeks, I focused on refining both the Analytics and Chatbot modules while completing the development of the AnalyticsML page. The Analytics page was reorganized to present visual charts and insights before the ranking table, making the data interpretation flow more intuitive. On the Chat page, I resolved persistent scrolling issues that caused the interface to jump when messages were sent by implementing a responsive auto-scroll mechanism that adjusts dynamically to chat content. The Chatbot was further improved to generate dataset-specific responses aligned with the `agg.csv` dataset through better structured prompt templates and contextual parameterization. Major milestones included the successful completion of the four AnalyticsML tasks: Forecasting, Anomaly Detection, Prediction, and Clustering, which now provide deeper analytical insight into sustainability metrics. Several challenges were encountered, including component overlapping between charts and tables on smaller screens, model response mismatches in dataset queries, and latency in rendering ML visualizations. These were resolved by refining Tailwind CSS breakpoints, restructuring data state management to reduce render cycles, and optimizing the ML calculation logic for smoother chart updates. Finally, the AnalyticsML page was fully separated from the Analytics page, added as a distinct header link, and tested for responsiveness and navigation integrity. This phase enhanced overall usability, data reliability, and technical stability of the GreenPath application, positioning it for the upcoming deployment milestone.

Repo Check in of Implementation Completed

The files/folders I have checked in the repo since report 2 are as follows:

Recent committed files under `Implementation/greenpath/` include:

`src/pages/Analytics.tsx` – Rearranged layout placing visualizations above ranking table, improved rendering and responsiveness.

`src/pages/Chat.tsx` – Fixed scroll mode, implemented responsive chat adjustment, aligned Chatbot responses with `agg.csv` dataset.

`src/pages/AnalyticsML.tsx` – Implemented forecasting, anomaly detection, prediction, and clustering modules, finalized separation as standalone page.

`src/layouts/RootLayout.tsx` – Added Analytics ML as distinct header link with consistent navigation styling.

`src/router.tsx` – Updated routing for AnalyticsML page integration.

`src/index.css` – Tailwind CSS and spacing adjustments to unify UI design.